Poorly distributed polynomials over finite fields are low-rank

Based on the paper by B. Green and T. Tao (2007)

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Polynomials over finite field

We fix the finite field $\mathbb{F}$ of prime order

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When $\mathrm{char}(\mathbb{F}) > d$ the class $\mathcal{P}_d$ can be defined differently: the function $P\colon \mathbb{F}^n \to \mathbb{F}$ is a polynomial if

$$D_{h_1} \cdots D_{h_{d+1}} P \equiv 0, \qquad h_1, \ldots, h_{d+1} \in \mathbb{F}^n.$$

Here $D_h P(x) = P(x+h) - P(x)$ denotes the discrete derivative in direction $h \in \mathbb{F}^n$.

Polynomial rank

Degree d polynomial rank

For function $P\colon \mathbb{F}^n \to \mathbb{F}$ we define $\operatorname{rank}_d(P)$ as the smallest k such that

$$P(x) = B\big(Q_1(x), \dots, Q_k(x)\big), \qquad Q_1, \dots, Q_k \in \mathcal{P}_d.$$

Equivalently, P is measurable with respect to $\sigma(Q_1, \dots, Q_k)$.

If $d = 0$ then $\operatorname{rank}_0(P) = 0$ if P is constant and $\operatorname{rank}_0(P) = \infty$ otherwise.

When $d \geq 1$ we always have $\operatorname{rank}_d(P) \leq n < \infty$.

Lack of equidistribution implies bounded rank

We define $e_{\mathbb{F}} \colon \mathbb{F} \to \mathbb{C}$ *as*

$$e_{\mathbb{F}}(t) = \exp(2\pi i t/|\mathbb{F}|)$$

and, for function $P \colon \mathbb{F}^n \to \mathbb{F}$ *, introduce the exponential bias*

$$\mathbb{E}_{x \in \mathbb{F}^n} e_{\mathbb{F}}(P(x)) = \frac{1}{|\mathbb{F}|^n} \sum_{x \in \mathbb{F}^n} e_{\mathbb{F}}(P(x)).$$

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Theorem 1.8

Suppose that d satisfies $0 \leq d < |\mathbb{F}|$. If $P \in \mathcal{P}_d(\mathbb{F}^n)$ is such that

$$|\mathbb{E}_{x \in \mathbb{F}^n} e_{\mathbb{F}}(P(x))| \geq \delta.$$

Then it satisfies

$$\mathrm{rank}_{d-1}(P) \leq C(\mathbb{F}, \delta, d).$$

Small d examples

If $P \in \mathcal{P}_1(\mathbb{F}^n)$ *is linear, then*

$$\begin{cases} \left|\mathbb{E}_{x \in \mathbb{F}^n} e_{\mathbb{F}}(P(x))\right| = 1, & \operatorname{rank}_0(P) = 0, \\ \mathbb{E}_{x \in \mathbb{F}^n} e_{\mathbb{F}}(P(x)) = 0, & \operatorname{rank}_0(P) \neq 0. \end{cases}$$

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Proposition

Suppose that $P \in \mathcal{P}_2(\mathbb{F}^n)$ satisfies $|\mathbb{E}_{x \in \mathbb{F}^n} e_{\mathbb{F}}(P(x))| \geq \delta$ then

$$\operatorname{rank}_1(P) \ll \log_{|F|} \frac{1}{\delta}.$$

Equivalently, this rewrites as

$$|\mathbb{E}_{x \in \mathbb{F}^n} e_{\mathbb{F}}(P(x))| \ll |F|^{-c \operatorname{rank}_1(P)}.$$

Exponential bias for quadratic polynomials

Proposition

Suppose that $P \in \mathcal{P}_2(\mathbb{F}^n)$ *satisfies* $|\mathbb{E}_{x\in\mathbb{F}^n} e_{\mathbb{F}}(P(x))| \geq \delta$ *then*

$$\mathrm{rank}_1(P) \ll \log_{|F|} \frac{1}{\delta}.$$

Proof.

$$\delta^2 \leq |\mathbb{E}_{x\in\mathbb{F}^n} e_{\mathbb{F}}(P(x))|^2 = \mathbb{E}_{x,y\in\mathbb{F}^n} e_{\mathbb{F}}(P(x)-P(y))$$
$$= \mathbb{E}_{x,h\in\mathbb{F}^n} e_{\mathbb{F}}(D_hP(x)) = \mathbb{P}_{h\in\mathbb{F}^n}(D_hP \in \mathcal{P}_0(\mathbb{F}^n)).$$

Rewrite P in the form $P(x) = \sum_{i,j} a_{i,j}x_ix_j + L(x)$ with $L \in \mathcal{P}_1(\mathbb{F}^n)$,

$$P(x) = B(x,x) + L, \qquad B(x,y) = \sum_{i,j} \frac{1}{2} a_{i,j}x_iy_j + \sum_{i,j} \frac{1}{2} a_{i,j}x_jy_i.$$

We see that $D_hP(x) - 2B(x,h) \in \mathcal{P}_0(\mathbb{F}^n)$ hence

$$\mathbb{P}_{h\in\mathbb{F}^n}(B(h,x) = 0 \text{ for all } x \in \mathbb{F}^n) \geq \delta^2.$$

Suitable h form a linear subspace of codimension at most $O(\log_{|F|} \frac{1}{\delta})$. □

Factor definition

Factor of degree d

By a *factor* of degree d on $\mathbb{F}^n$ we mean a collection of polynomials

$$\mathcal{F} = (P_{i,j})_{1 \le i \le d,\, 1 \le j \le M_i}, \qquad P_{i,j} \in \mathcal{P}_i(\mathbb{F}^n).$$

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It will be convenient for think of $\mathcal{F}$ in terms of the sigma algebra

$$\sigma(\mathcal{F}) = \sigma((P_{i,j})_{i,j}).$$

We write $\Sigma = \mathbb{F}^{M_1} \times \cdots \times \mathbb{F}^{M_d}$ and call this the *configuration space* of $\mathcal{F}$.
We write $\Phi : \mathbb{F}^n \to \Sigma$ for the *evaluation map* given by

$$\Phi(x) = (P_{i,j}(x))_{1 \le i \le d,\, 1 \le j \le M_i}.$$

Regularity of a factor

Let $\mathcal{F}$ be a factor of order d, and let $F\colon \mathbb{Z}_+ \to \mathbb{R}_+$ be a growth function. We say that $\mathcal{F}$ is *F-regular* if

$$\operatorname{rank}_{i-1}\left(\sum_{j=1}^{M_i} c_{i,j} P_{i,j}\right) \geq F\big(\dim(\mathcal{F})\big)$$

for all $1 \leq i \leq d$ and all coefficients $c_{i,1}, \ldots, c_{i,M_i} \in \mathbb{F}$ that are not all zero.

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Lemma

Given a degree-d factor $\mathcal{F}$ and a growth function F, there is an F-regular extension $\mathcal{F}'$ of degree d with

$$\dim \mathcal{F}' \leq C(F, d, \dim \mathcal{F}).$$

The factor regularity lemma

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Proof.

We induct on the dimension vector $(M_1, \ldots, M_d)$. If regularity does not hold then for some $1 \leq i \leq d$ there exist $c_1, \ldots, c_{M_i}$ such that

$$N := \mathrm{rank}_{i-1}(\sum c_j P_{i,j}) < F(\dim(\mathcal{F})),$$

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Delete $P_{i,1}$ from $\mathcal{F}_i$ and insert $Q_1, \ldots, Q_N$ into $\mathcal{F}_{i-1}$. We have

$$\sigma(\mathcal{F}) \subset \sigma(\mathcal{F}').$$

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The new dimension vector is

$$(M_1, \ldots M_{i-2}, M_{i-1} + N, M_i - 1, M_{i+1}, \ldots, M_d).$$



Ingredients of the proof of the main theorem

Lemma 3.1, Bogdanov – Viola lemma

If $P \in \mathcal{P}_d(\mathbb{F}^n)$ and $|\mathbb{E}_{x \in \mathbb{F}^n} e_{\mathbb{F}}(P(x))| \geq \delta$, then for any $0 < \sigma \leq 1$ there is a function $\tilde{P}\colon \mathbb{F}^n \to \mathbb{F}$ such that

$$\mathrm{rank}_{d-1}(\tilde{P}) \leq \frac{|\mathbb{F}|^5}{\delta^2 \sigma}, \qquad \mathbb{P}_{x \in \mathbb{F}^n}\Big(P(x) \neq \widetilde{P}(x)\Big) \leq \sigma.$$

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Proposition 5.1

Suppose that Theorem 1.8 holds for all degrees up to $d-1$ and let $\sigma_d > 0$ be a small quantity to be specified later.

Let $P \in \mathcal{P}_d(\mathbb{F}^n)$ be a polynomial, F be rapidly growing growth function and $\mathcal{F}$ be an F-regular factor of degree $d-1$.

If $\widetilde{P} : \mathbb{F}^n \to \mathbb{F}$ is an $\mathcal{F}$-measurable function and

$$\mathbb{P}\big(P(x) = \widetilde{P}(x)\big) \geq 1 - \sigma_d,$$

then P is itself $\mathcal{F}$-measurable.

Proof of the main result

Theorem 1.8

Suppose that d satisfies $0 \leq d < |\mathbb{F}|$. If $P \in \mathcal{P}_d(\mathbb{F}^n)$ is such that

$$|\mathbb{E}_{x \in \mathbb{F}^n} e_{\mathbb{F}}(P(x))| \geq \delta.$$

Then it satisfies $\mathrm{rank}_{d-1}(P) \leq C(\mathbb{F}, \delta, d)$.

Proof.

We induct on d. Take the function $\widetilde{P}$ from Lemma 3.1:

$$N := \mathrm{rank}_{d-1}(\tilde{P}) \leq |\mathbb{F}|^5/(\delta^2 \sigma_d), \qquad \mathbb{P}_{x \in \mathbb{F}^n}\left(P(x) \neq \widetilde{P}(x)\right) \leq \sigma_d,$$

$$\widetilde{P} = B(Q_1, \ldots, Q_N), \qquad \mathcal{F} = (Q_j)_{1 \leq j \leq N}.$$

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The result follows upon applying Proposition 5.1: P is measurable with respect to $\sigma(\mathcal{F}')$ hence $\mathrm{rank}_{d-1}(P) \leq \dim \mathcal{F}'$.

Proof of Bogdanov–Viola lemma

Lemma 3.1, Bogdanov – Viola lemma

If $P \in \mathcal{P}_d(\mathbb{F}^n)$ and $|\mathbb{E}_{x \in \mathbb{F}^n} e_{\mathbb{F}}(P(x))| \geq \delta$, then for any $0 < \sigma \leq 1$ there is a function $\tilde{P}\colon \mathbb{F}^n \to \mathbb{F}$ such that

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Proof.

We will construct $\widetilde{P}$ measurable with respect to

$$\sigma(D_{h_1}P, \ldots, D_{h_N}P), \qquad h_1, \ldots, h_N \in \mathbb{F}^n.$$

For each $r \in \mathbb{F}$, define a measure μ_r on $\mathbb{F}$,

$$\mu_r(t) = \mathbb{P}_{x\in\mathbb{F}^n}(P(x) = t + r).$$

Fix $x \in \mathbb{F}^n$. We have

$$\mathbb{P}_h(D_hP(x) = t) = \mathbb{P}_h(P(x+h) = t + P(x)) = \mu_{P(x)}(t).$$

Sample $h_1, \ldots, h_N$ randomly and define $\widetilde{P}(x) = r$, where r minimizes

$$\|\frac{1}{N}\sum_{i=1}^{N} \delta_{D_{h_i}P(x)} - \mu_r\|.$$

Measures are separated

Proof.

Recall that μ_r is defined as

$$\mu_r(t) = \mathbb{P}_{x \in \mathbb{F}^n}(P(x) = t + r).$$

Assertion $|\mathbb{E}_{x \in \mathbb{F}^n} e_{\mathbb{F}}(P(x))| \geq \delta$ rewrites as $|\sum_{t \in \mathbb{F}} e_{\mathbb{F}}(t)\mu_0(t)| \geq \delta$.

For every $d \in \mathbb{F}$ we have

$$\mu_0(t) = \mathbb{P}_{x \in \mathbb{F}^n}(P(x) = t) = \mathbb{P}_{x \in \mathbb{F}^n}(P(x) = t - d + d) = \mu_d(t - d),$$

$$\sum_{t \in \mathbb{F}} e_{\mathbb{F}}(t)\mu_0(t) = e_{\mathbb{F}}(d) \sum_{t \in \mathbb{F}} e_{\mathbb{F}}(t - d)\mu_0(t) = e_{\mathbb{F}}(d) \sum_{t \in F} e_{\mathbb{F}}(t)\mu_d(t),$$

$$(1 - e_{\mathbb{F}}(d)) \sum_{t \in \mathbb{F}} e_{\mathbb{F}}(t)\mu_0(t) = e_{\mathbb{F}}(d) \sum_{t \in \mathbb{F}} e_{\mathbb{F}}(t)(\mu_d(t) - \mu_0(t)).$$

For $d \neq 0$ we get

$$\|\mu_0 - \mu_d\| := \sum_t |\mu_0(t) - \mu_d(t)| \geq |1 - e_{\mathbb{F}}(d)| \left| \sum_t e_{\mathbb{F}}(t)\mu_0(t) \right| \geq 4\delta/|\mathbb{F}|.$$

By translation invariance we conclude that

$$\|\mu_r - \mu_s\| \geq 4\delta/|\mathbb{F}|, \qquad r \neq s.$$

End of the proof

Proof.

Sample $h_1, \dots, h_N$ randomly and define $\widetilde{P}(x) = r$, where r minimizes

$$\|\mu_{obs} - \mu_r\|, \qquad \mu_{obs} = \frac{1}{N}\sum_{i=1}^{N} \delta_{D_{h_i}P(x)}.$$

The measures m_r are separated

$$\|\mu_r - \mu_s\| \geq 4\delta/|\mathbb{F}|, \qquad r \neq s.$$

If $\widetilde{P}(x) \neq P(x)$ then $\|\mu_{obs} - \mu_{P(x)}\| \geq 2\delta/|\mathbb{F}|$. We claim that

$$\mathbb{P}_{h_1,\dots,h_N}\left(\|\mu_{obs} - \mu_{P(x)}\| \geq 2\delta/|\mathbb{F}|\right) \leq \frac{\sigma}{|\mathbb{F}|}.$$

It suffices to prove that for every $x, t \in \mathbb{F}$

$$\mathbb{P}_{h_1,\dots,h_N}\left(|\mu_{obs}(t) - \mu_{P(x)}(t)| \geq 2\delta/|\mathbb{F}|^2\right) \leq \frac{\sigma}{|\mathbb{F}|}.$$

By the definition we have $\mu_{obs}(t) = \frac{1}{N}\sum_{i=1}^{N} \delta_{D_{h_i}P(x)}(t)$. The final step is Chebyshev's inequality

$$\mathbb{P}\left(\left|\frac{Y_1 + \dots + Y_N}{N} - \overline{Y}\right| \geq \eta\right) \leq \frac{1}{4N\eta^2}.$$

Proof of the rigidity theorem in the simplest case

Proposition 5.1

Suppose that Theorem 1.8 holds for all degrees up to $d-1$ and let $\sigma_d > 0$ be a small quantity to be specified later.

Let $P \in \mathcal{P}_d(\mathbb{F}^n)$ be a polynomial, F be rapidly growing growth function and $\mathcal{F}$ be an F-regular factor of degree $d-1$.

If $\widetilde{P} : \mathbb{F}^n \to \mathbb{F}$ is an $\mathcal{F}$-measurable function and that

$$\mathbb{P}\big(P(x) = \widetilde{P}(x)\big) \geq 1 - \sigma_d,$$

then P is itself $\mathcal{F}$-measurable.

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Lemma (Non-zero polynomials do not vanish almost everywhere)

If $P \in \mathcal{P}_d(\mathbb{F}^n)$ is such that $\mathbb{P}_{x\in\mathbb{F}^n}(P(x) = 0) > 1 - 2^{-d}$ then $P \equiv 0$.

Proof.

We proceed by induction on d. For any fixed h we have

$$\mathbb{P}_{x\in\mathbb{F}^n}(P(x+h) = P(x) = 0) > 1 - 2^{-(d-1)}.$$

The inductive hypothesis gives $P(x+h) - P(x) \equiv 0$. □

Counting the atoms

Recall that for a factor $\mathcal{F} = (P_{i,j})_{1 \le i \le d, 1 \le j \le M_i}$ *we write* $\Sigma = \mathbb{F}^{M_1} \times \cdots \times \mathbb{F}^{M_d}$ *for configuration space of* $\mathcal{F}$. $\Phi : \mathbb{F}^n \to \Sigma$ *for the evaluation map given by*

$$\Phi(x) = (P_{i,j}(x))_{1 \le i \le d,\ 1 \le j \le M_i}.$$

The atoms of the $\sigma(\mathcal{F})$ *have the form*

$$\{x \in \mathbb{F}^n : \Phi(x) = t\}, \qquad t \in \Sigma.$$

Lemma

Let $d \ge 1$, and $\varepsilon > 0$. Suppose that Theorem 1.8 is true for degrees up to d. Then there exists a growth function F (depending on d and ε) such that if $\mathcal{F}$ is an F-regular factor of degree d on $\mathbb{F}^n$ then we have the estimate

$$\mathbb{P}_{x \in \mathbb{F}^n}(\Phi(x) = t) = (1 + O(\varepsilon))\frac{1}{|\Sigma|}, \qquad t \in \Sigma.$$

Counting the atoms

Proof.

We may expand the condition $\Phi(x) = t$ in a form

$$\mathbb{P}_x(\Phi(x) = t) = \frac{1}{|\Sigma|}\sum_{r\in\Sigma}\mathbb{E}_{x\in\mathbb{F}^n} e_{\mathbb{F}}\big(r\cdot(\Phi(x)-t)\big)$$

$$= \frac{1}{|\Sigma|} + \frac{1}{|\Sigma|}\sum_{r\in\Sigma\setminus\{0\}}\mathbb{E}_{x\in\mathbb{F}^n} e_{\mathbb{F}}\big(r\cdot(\Phi(x)-t)\big).$$

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For every $r \in \Sigma$ we have

$$r\cdot\Phi(x) = \sum_{i=1}^{d}\sum_{j=1}^{M_j} r_{i,j}P_{i,j}(x) = \sum_{i=1}^{d} Q_i, \qquad Q_i \in \mathrm{Span}(\mathcal{F}_i).$$

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$$\mathbb{P}_x(\Phi(x) = t) = \frac{1}{|\Sigma|} \sum_{r \in \Sigma} \mathbb{E}_{x \in \mathbb{F}^n} e_{\mathbb{F}}\big(r \cdot (\Phi(x) - t)\big)$$

$$= \frac{1}{|\Sigma|} + \frac{1}{|\Sigma|} \sum_{r \in \Sigma \setminus \{0\}} \mathbb{E}_{x \in \mathbb{F}^n} e_{\mathbb{F}}\big(r \cdot (\Phi(x) - t)\big).$$

For every $r \in \Sigma$ we have

$$r \cdot \Phi(x) = \sum_{i=1}^{d} \sum_{j=1}^{M_j} r_{i,j} P_{i,j}(x) = \sum_{i=1}^{d} Q_i, \qquad Q_i \in \mathrm{Span}(\mathcal{F}_i).$$

It therefore suffices to show that

$$\mathbb{E}_{x \in \mathbb{F}^n} e_{\mathbb{F}} \left(\sum_{i=1}^{d} Q_i \right) = O\left(\varepsilon / |\mathbb{F}|^{\dim(\Sigma)} \right), \qquad Q_i \in \mathrm{Span}(\mathcal{F}_i).$$

Counting the atoms

Proof.

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Let s be such that not all $r_{s,j}$ are zero. As $\mathcal{F}$ is F-regular, we have

$$\mathrm{rank}_{s-1}\left(\sum_{i=1}^{s} Q_i\right) \geq \mathrm{rank}_{s-1} Q_s - 1 \geq F(\dim(\mathcal{F})) - 1.$$

Good atoms

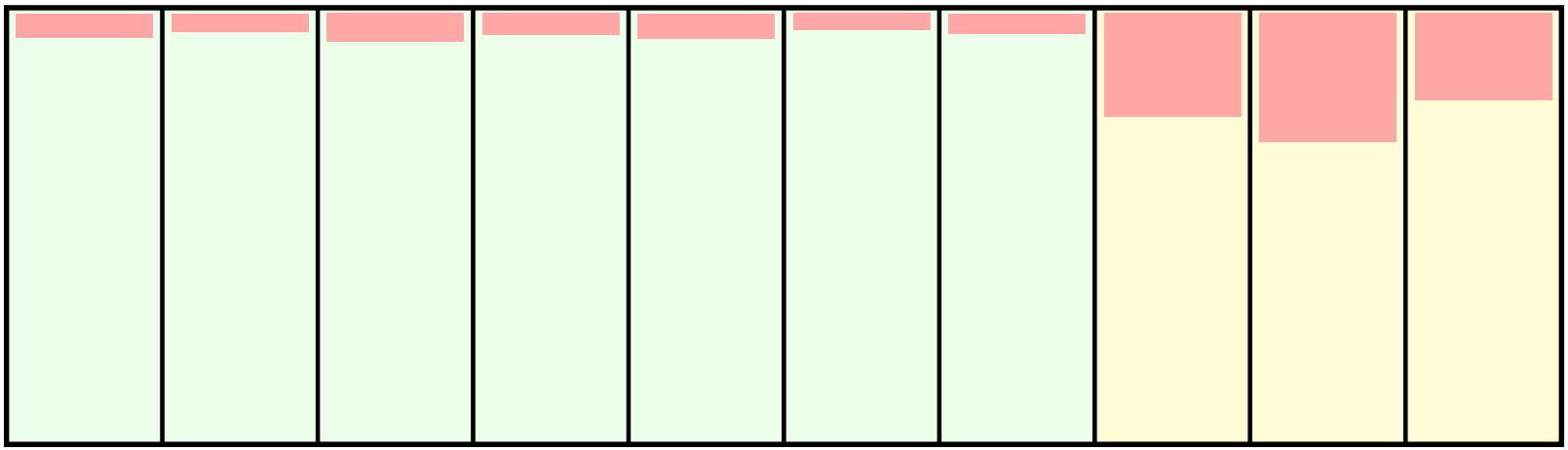
The assertion of the proposition states

$$\mathbb{P}_{x \in \mathbb{F}^n}\left(P(x) = \widetilde{P}(x)\right) \geq 1 - \sigma_d.$$

We call an atom A of $\mathcal{F}$ *good* if

$$\mathbb{P}_{x \in A}\left(P(x) = \widetilde{P}(x)\right) \geq 1 - \sqrt{\sigma_d}.$$

From the counting lemma we get that $1 - \sqrt{\sigma_d}$ of atoms are good.



Good atoms

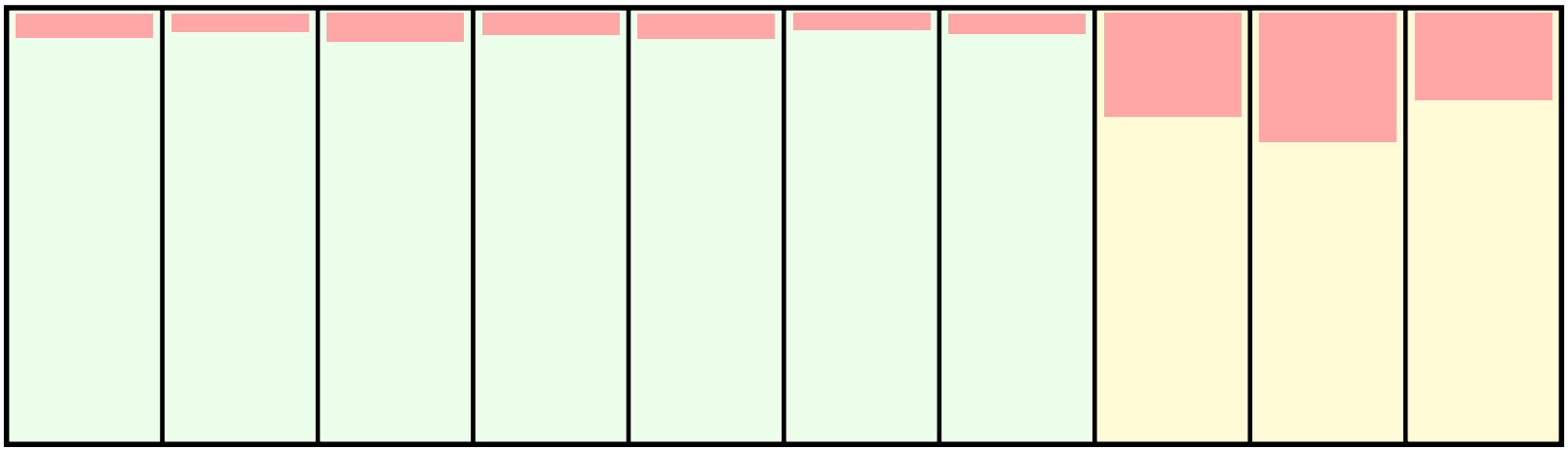
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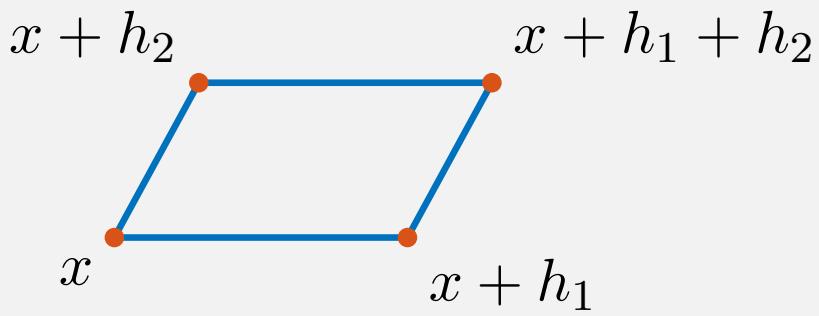
- *All good blocks are perfect.*
- *Every good block is actually perfect too.*

Polynomials and parallelepipeds

If $P \in \mathcal{P}_d(\mathbb{F}^n)$ *then for every* $h \in (\mathbb{F}^n)^{d+1}$ *we have*

$$\sum_{\omega \in \{0,1\}^{d+1}} (-1)^{|\omega|} P(x + \omega \cdot h) = D_{h_1} \dots D_{h_{d+1}} P(x) = 0.$$

For $P \in \mathcal{P}_1(\mathbb{F}^n)$ *the alternating sum around these four points is zero:*

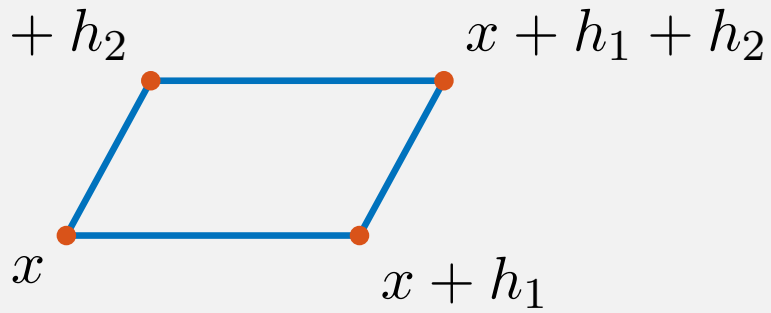


Polynomials and parallelepipeds

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For $P \in \mathcal{P}_1(\mathbb{F}^n)$ *the alternating sum around these four points is zero:*



If P *and* $\widetilde{P}$ *coincide on* $x + w \cdot h$ *for all* $w \neq 0^{d+1}$ *then also* $P(x) = \widetilde{P}(x)$.

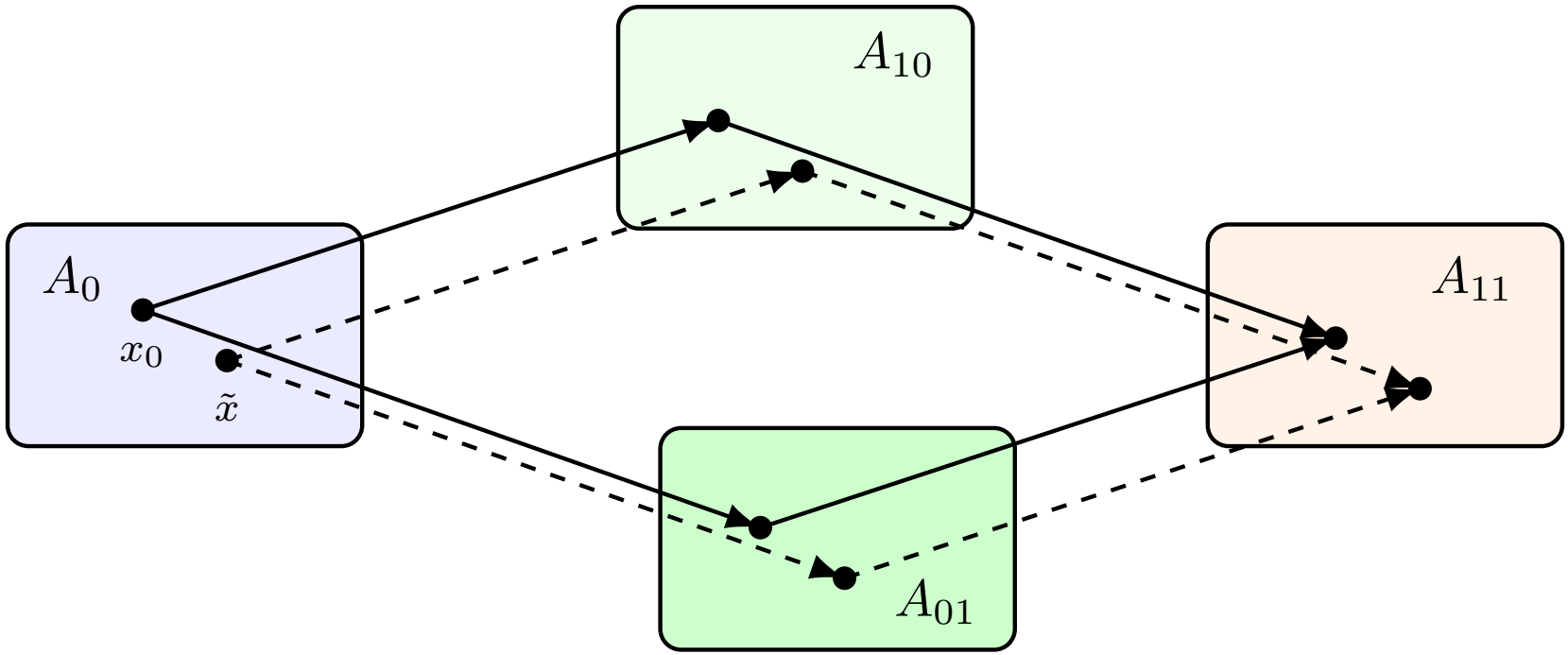
More counting

Every parallelepiped $\{x + w \cdot h\}_{w \in \{0,1\}^{d+1}}$ *defines a vector of atoms*

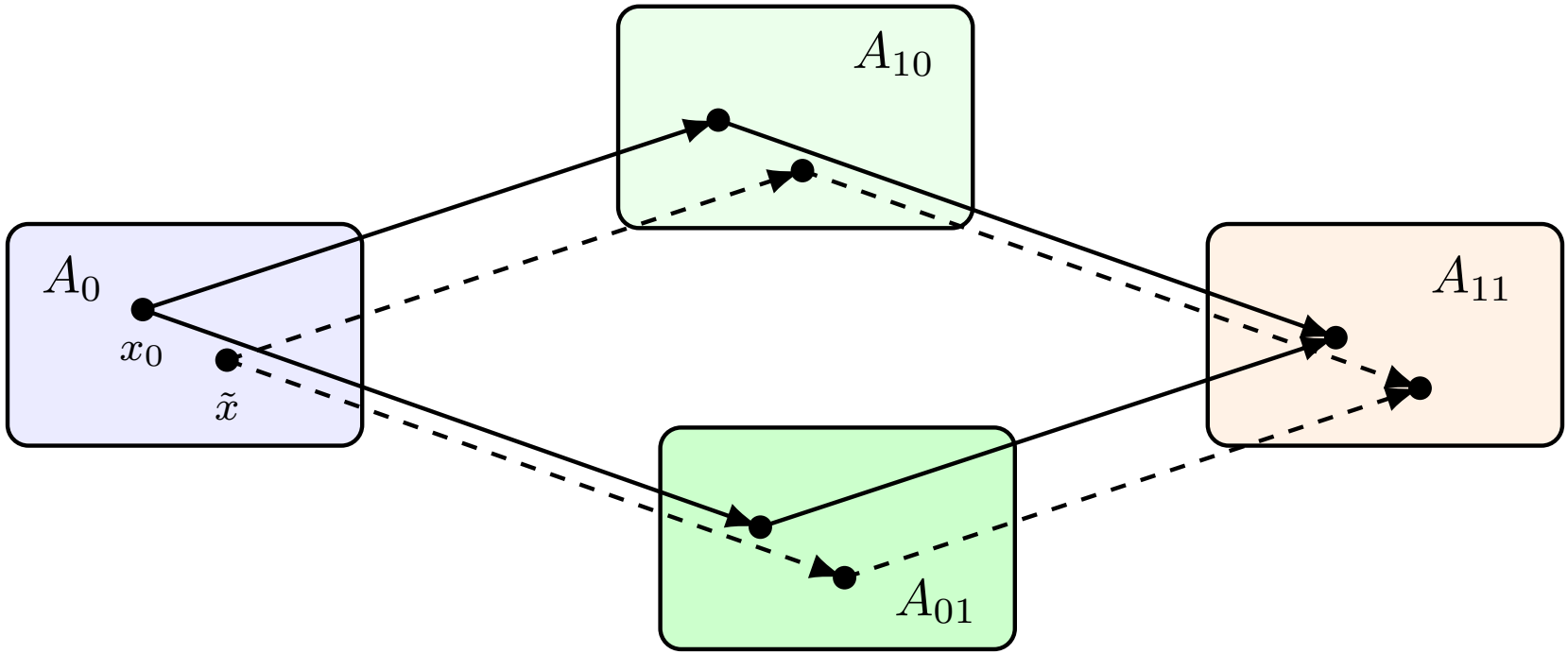
$$(A_w)_{w \in \{0,1\}^{d+1}}, \qquad x + w \cdot h \in A_w.$$

Parallelepiped counting

If there is a parallelepiped $x_0 + w \cdot h$ that corresponds to (A_w) then for every $\tilde{x} \in A_0$ there exist a lot of such parallelepipeds starting at $\tilde{x}$.



Parallelepipeds inside the atoms



Taking $h = 0$ we deduce that there are a lot of parallelepipeds inside every atom.

Perfect atoms

Lemma

Let A be a good atom of $\mathcal{F}$. For every $x \in A$ there is a tuple h such that $x_w = x + w \cdot h \in A$ and $P(x_w) = \widetilde{P}(x_w)$ for all $w \in \{0,1\}^{d+1} \setminus 0^{d+1}$.

Perfect atoms

Lemma

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If $P \in \mathcal{P}_d(\mathbb{F}^n)$ then for every $h \in (\mathbb{F}^n)^{d+1}$ we have

$$\sum_{\omega \in \{0,1\}^{d+1}} (-1)^{|\omega|} P(x + \omega \cdot h) = 0$$

Proposition

If A is a good atom then $P \equiv \widetilde{P}$ on A.

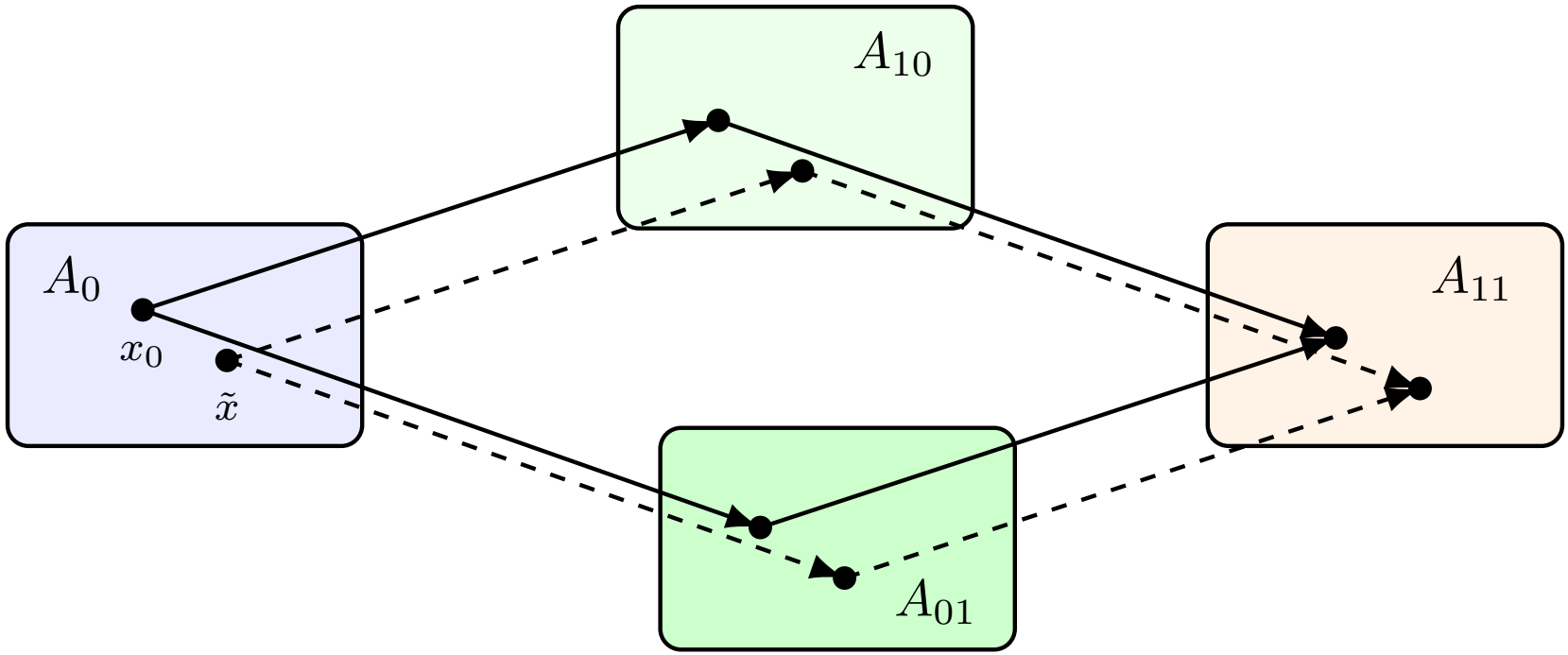
Proof.

Fix $x \in A$ and let h be a tuple from the lemma. We have

$$P(x) = -\sum_{w \neq 0} (-1)^{|w|} P(x + w \cdot h) = -\sum_{w \neq 0} (-1)^{|w|} \widetilde{P}(x + w \cdot h) = \widetilde{P}(x).$$

□

Processing bad atoms



We need to find a single parallelepiped $\{x + w \cdot h\}$ *such that* A_w *is a good block for every* $w \neq 0$. *Indeed,*

$$P(x) = -\sum_{w \neq 0} (-1)^{|w|} P(x + w \cdot h) = -\sum_{w \neq 0} (-1)^{|w|} c_w.$$

Parallelepiped with vertices in good blocks

We choose points $h_1, \dots, h_{d+1}$ randomly.

For $\omega \neq 0^{d+1}$, the probability that $x + \omega \cdot h$ lies in a bad atom is $O(\sqrt{\sigma_d})$. There are 2^{d+1} different elements w hence the assertion $\sigma_d < c/2^{2d}$ for some small $c > 0$ shows the existence of the choice of h such that every $x + w \cdot h$ belongs to the good atom.

Proof scheme of the main theorem

